Steven Kolawole

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Research Focus

The efficiency of ML inference: model compression, KV cache optimization, adaptive serving (cascading and speculative decoding), and inference parallelism for production systems.

Education

Ph.D. in Computer Science, Carnegie Mellon University 2023 – 2028 (exp.)

Language Technologies Institute, School of Computer Science. Advisor: Prof. Virginia Smith.

B.Sc. in Computer Science, Federal University of Agriculture, Abeokuta 2017 – 2023

Research Experience

Applied Scientist Intern, AWS AGI Foundations May – Aug 2026

San Francisco, CA. Inference optimization science team (Bedrock); mentors: Drs. Nathan Pemberton & Kyle Ulrich. In-place speculative decoding for long-context inference, drafting from the target model's own verifying pass with no separate draft model.

Graduate Student Researcher, Carnegie Mellon University Aug 2023 – Present

Pittsburgh, PA. Advised by Prof. Virginia Smith. Algorithmic methods for efficient ML inference.

Independent Researcher, ML Collective Apr 2021 – Present

Remote. Pre-Ph.D. research mentored by Drs. Rosanne Liu & Jason Yosinski; now mentoring junior researchers within the same open-science pipeline. Resource-constrained NLP, federated learning, and efficient ML.

Publications

* *denotes equal contribution.*

- **S. Kolawole**, K. Santhanam, V. Smith, P. Thaker. *ParallelPrompt: Extracting Parallelism from Large Language Model Queries.* NeurIPS 2025, Datasets & Benchmarks Track. OpenReview · arXiv:2506.18728.
- D. Soiffer, **S. Kolawole**, V. Smith. *Semantic Agreement Enables Efficient Open-Ended LLM Cascades.* EMNLP 2025, Industry Track. ACL Anthology · arXiv:2509.21837.
- **S. Kolawole***, D. Dennis*, A. Talwalkar, V. Smith. *Agreement-Based Cascading for Efficient Inference.* Transactions on Machine Learning Research (TMLR), 2025. OpenReview · arXiv:2407.02348.
- **S. Kolawole***, L. Dery*, J.-F. Kagy, V. Smith, G. Neubig, A. Talwalkar. *Everybody Prune Now: Structured Pruning of LLMs with only Forward Passes* (Bonsai). arXiv:2402.05406.
- **S. Kolawole**, O. Osakuade, N. Saxena, B. K. Olorisade. *Sign-to-Speech Model for Sign Language Understanding: A Case Study of Nigerian Sign Language.* IJCAI 2022, Demo Track. Proceedings · arXiv:2111.00995.

Manuscripts in Review

- **S. Kolawole**, J. Campos Salas, N. Pemberton, M. Kleindessner, A. Khetan, K. Ulrich. *Janus: Speculative Decoding Inside the Verifying Pass.* Under review, MLSys 2027.
- **S. Kolawole**, P. Jim, O. M. Busoye, G. Bagai, V. Smith. *A Tilted Bowl Is Not a Slippery Slope: Compressing Looped Models.* Under review, ICLR 2027. arXiv:2609.39277.

- **S. Kolawole**, V. Smith. *EpiKV: Epiphany-Aware KV Cache Eviction Without the Attention Matrix.* Under review, ICLR 2027. arXiv:2606.26472.
- P. Jim*, **S. Kolawole***, O. M. Busoye, G. Bagai, V. Smith. *What Survives When You Compress a Recursive Reasoner for the Edge?* arXiv:2606.26488.

Workshop Papers

- M. Oduwole, . . . , A. Owodunni, **S. Kolawole**. *From Scarcity to Efficiency: Investigating the Effects of Data Augmentation on African Machine Translation.* LM4UC @ AAAI 2026 (Springer CCIS). OpenReview. **[senior/last author]**
- N. Obiefuna, S. Oyeneye*, S. Odunaiya*, I. Oyelaja*, **S. Kolawole**. *Privacy Isn't Free: Benchmarking the Systems Cost of Privacy-Preserving ML.* ES-FoMo @ ICML 2025. OpenReview. **[senior/last author]**
- B. Awobade*, M. Oduwole*, **S. Kolawole**. *What Happens When Small Is Made Smaller? Exploring the Impact of Compression on Small Data Pretrained Language Models.* AfricaNLP @ ICLR 2024. arXiv:2404.04759. **[senior/last author]**
- C. Leong, . . . , **S. Kolawole**, et al. *Adapting to the Low-Resource Double-Bind: Investigating Low-Compute Methods on Low-Resource African Languages.* AfricaNLP @ ICLR 2023. arXiv:2303.16985.
- N. Alam*, **S. Kolawole***, S. Sethi*, N. Bansali, K. Nguyen. *Vision Transformers for Mobile Applications: A Short Survey.* arXiv, 2023. arXiv:2305.19365.

Peer Review & Professional Service

- **ML/AI:** NeurIPS (2024, 2026); ICML (2026); ICLR (2025, 2027); AISTATS (2025, 2026); AAAI (2026); ES-FoMo @ ICML (2024).
- **NLP:** COLM (2025, 2026); ACL Demo (2025, 2026); EMNLP Demo (2024, 2025, 2026).

Honors & Awards

- **Research Grant, OpenMined** (Ensembles & Routing): extending cascading to frontier models. 2026
- **Scholar, Machine Learning Summer School (MLSS) NYC** (Columbia): competitively selected. 2026
- **Research Grant, Algorand Foundation**: $114,635 Ecosystem grant for an ML research-engineering project (the ASAlytics platform). 2022
- **National AI Champion**, Nigeria Computer Society AI Summit: first position, AI Challenge. 2022
- **Ideathon Winner** ($5,000 Google Cloud credits), Deep Learning Indaba, the continent's foremost ML gathering. 2022
- **Best Poster, Data Science Nigeria.** 2021
- **Winner, DeepQuest AI Challenge.** 2021
- **Scholar, MTN Foundation Science & Technology Scholarship**: selection rate 1.8%. 2020, 2021
- **Led AI+ Abeokuta to Data Science Nigeria's "AI City of the Year"**: as community lead, produced the largest national-bootcamp cohort of any Nigerian city. 2020
- **Runner-up, "Mr. Algorithm," Data Science Nigeria**: nationwide community vote, for technical excellence and knowledge-sharing. 2020
- **Fellow, Young Professionals Bootcamp**: selection rate 3.6%. 2020
- **Gold Medal, NaijaHacks**: AgroAI plant-disease-detection app; top 12 of 376 teams. 2019

Leadership & Community Service

- **Founder & Lead, ML Collective Africa** 2023 – Present
  Founded and lead ML Collective's only regional chapter, 300+ members across dozens of research groups. Set the research agenda, supervise the groups and advise their projects, and raised $50,000+ to send African researchers to the Deep Learning Indaba. Members' papers have won best paper

awards at MICCAI 2026's AFRICAI workshop and AfricaNLP 2025; chaired and secured sponsorships for its Indaba 2026 workshop.

- **Campus Lead, AI+ FUNAAB; lead, AI+ Abeokuta** (Data Science Nigeria) 2019 – 2021
  Built a structured remote preparation program that took the community from five national qualifiers (2019) to 38 of 42 entrants selected (2020; 28 in the national top 250, nine in the national top 20).
- **Community Lead, Google Developer Student Clubs, FUNAAB** 2021 – 2022
  Technical training reaching 3,000+ students; 600+ taught personally.
- **Mentor, STEM for Development** 2023 – 2025
  Mentor 20+ graduate-school applicants per year toward U.S. and other leading research programs.
- **Mentor, She Code Africa** 2021 – 2022
  Mentored women entering machine learning and data science.

Selected Teaching

Guest Lecturer (Model Compression) & Teaching Assistant, 10-718 Machine Learning in Practice, CMU Spring 2026

Invited Lecturer, AI Saturdays Lagos (AI6) Fall 2023

Selected Invited Talks & Presentations

(Research orals marked †; invited talks/panels noted.)

- **2026:** "Doing ML Research from the Margins," skills session, Deep Learning Indaba (Lagos); "A SOTA Overview of Efficient AI, and What It Means for Africa," IndabaX Nigeria *(invited)*.
- **2025:** Black in AI ELAI graduate-school talk *(invited)*; CMU Africa Research Club *(invited, ×2)*.
- **2024:** Low-resource LLM practicals, Deep Learning Indaba (Senegal).
- **2023:** OSCAFEST.
- **2022:** IJCAI†; Cohere For AI (C4AI) Independent-Research panel *(invited)*; Deep Learning Indaba ML-efficiency talk (Tunisia); Lanfrica Talks *(invited)*; EuroPython (Dublin); PyConDE & PyData Berlin; Berlin Buzzwords; Build Stuff; SciPy (Austin, Texas); NCS AI Summit; DataFest Africa; GDG Abeokuta DevFest *(invited)*.
- **2021:** NeurIPS ML4D†; Masakhane Speaker Series *(invited)*; PyData Global; EuroPython; PyCon APAC; Conf42 Python; ML Collective Research Jam (×3).
- **2020:** PyData Global; PyCon Africa.

Selected Industry Experience

Founding ML Engineer, ASAlytics 2022

Backend systems for real-time opinion-mining/analytics of blockchain's digital assets; the vehicle for the Algorand Foundation research-engineering grant.

Data Science Intern, SeqHub Analytics LLC, New Haven, CT 2021

Fine-tuned mT5 for low-resource Nigerian-language machine translation; built microservices and a dockerized analytics dashboard.

Media: interview on the sign-to-speech research, AIhub (2022); talk recordings on PyVideo.

Updated November 2026